

Reverse Engineering – HOMEWORKS requests 2024

TEAMS ← → studies

homeworks ← → due dates

TEAM #	SURNAME	SURNAME	SURNAME	SURNAME	SURNAME	SURNAME	Assigned mission
	1	2	3	4	5	6	
1	Ceglie F.	Paniz A.	Paolo F.	Piemonte F.	Pilotto C.	Proserpio A.	Jason
2	Cataffo F.	Cesarini L.	Clerici M.	Cotti G.	Dal Passo A.	Di Rago D.	Giove
3	Aioldi C.	Bontemps D.	De Luca L.	De Marco E.	Perusini G.	Poverini V.	Jason
4	Paltani D.	Pacursa A.	Persenico F.	Piraccini M.	Provenzi D.	Portantiolo M.	Soho
5	C. Poli	P. Vanelli	D. Paternoster	A. Turcu	R. Vidari	M. Pareschi	Juno
6	Giroudiere P.	Coulmy A.	Sonnabend C.	Ternot M.	E. Forsenberg	A. Glanvik	Jason
7	De Amicis V.	Manucci P.	Di Carlo P.	Conforto V.	Di Chiara S.	Carretta V.	Juno
8	Bachini P.	Belletti S.	Bolsi P.	De Magistris M.	De Marco M.	Topuz D.	Soho
9	Parazzi L.	Pavia N.	Benericetti M.	Ruoppolo M.	Forleo V.	Campanello V.	Juno
10	Khelafa L.	Lee JJ.	Soule Y.	Aboul F.	Lobo A.	Longari R.	Jason
11	Dalone G.	Tessarollo A.	Tartaglia D.	Mascaretti M.	Berretti D.	De Carli M.	Giove
12	Borcea D.	Ashfaq H.	Basile E.	Marrone M.	Mondini U.	Chilá I.	Jason
13	Mirri P.	Grandinetti R.	Righi L.	Nuccio G.	Natali M.	Miccoli F.	Soho
14	Campana D.	Monai F.	Barbieri M.	Varli A.	Santoro E.	Campanino M.	Juno
15	Ancillotti G.	Barbiera A.	Cicchi F.	Grimaldi S.	Laghi D.	Volontè F.	Giove
16	Vurro M. A.	Lakrad A.	Ignesti S.	Elzeiny O.	Fraccastoro A.	Grignaffini R.	Soho
17	Binotto A.	Braga M.	Centorame C.	Bagnoli B.	Gentili G.	Galbiati L.	Giove
18	Zapparoli M.	Rossetti S.	Taglioretti G.	Rezzaghi A.	Mariani A.	Pastori C.	Soho
19	Mandolini C.	Ondeggia M.	Huang Y.	Mastrorosa A.	Gravina M.	Mancay Ramirez G.	Jason
20	Bustreo T.	Nicali A.	Le Coeur M.	Patureau R.	Arboleda B. A.	Gomez Sanchez C.	Giove
21	Bove C.	Giardini C.	Privitera G.	Bertrand A.	Torrente C. A.	Bax M.	Juno
22	Cannavò A.	Giorgini F.	Banaji-Seillier M.	Bourgeois C.	Cavalletti P.	Rossi L.	Jason
23	Dowell M.	Moretti F.	Cerletti E.	Marzo V.	Faro D.	Anastasio G.	Juno
24	Boccia A.	Cutino S.	D'Antonio A.	Harunor T.	Mastroleo M.	Meanti F.	Giove
25	Modelli S.	Juvara M.	Nicolucci Balocco	Orsenigo S.	Beretta B.	Separovic M. T.	Jason
26	Bergamaschi C.	Bellini D.	Biraghi G.	Cattaneo M.	Parvu S.	Sivasli A.	Juno
27	Frassinella L.	Massini A.	Scudier G.	Mensi Weingrill E.	Ahlborn M.	Alexandus J.	Juno
28	Rapacioli R.	Troni A.	Ferraro M.	Siviero L.	Xompero T.	Gebert H.	Soho
29	Conte C.	Fiume E.	Loffredo E.	Gallo E.	Gardioli F.	Accordini L.	Soho
30	Calegari C.	Cipriano E. F.	Carone S.	Arbelo Cabrera P.	Quinn B.	Palmerini V. P.	Juno
31	Christoforou A.	Sciutto G.	Taeggi F.	Tebaldi A.	Turotti B. A.	Vilhelem J.	Jason
32	Casiero A.	Del Monte C.	Ferrantino M.	Festa L.	Formica L.	Marotta A.	Giove
33	de Pinto A.	D'Annunzio V.	De Bellis Vitti M.	Bianco S.	Rizzi M.	Santalucia A.	Soho
34	Bezzi D.	Kumar A.	Ghatole P.	Conti M.	Nuzzolo Y.	Smith J.	Giove
35	Baj M.	Daverio E.	Fontana P.	Lorenzi E.	Monti R.	Oikawa S.	Soho

new	SSEO RE delivery timeline		
	Homework #	Delivery deadline	Topic
	1	March 15, 2024 - 23:30h	Obj, FUNC
	2	March 29, 2024 - 23:30h	MA, PS
	3	April 14, 2024 -23:30h	TTMTC
	4	April 28, 2024 -23:30h	ADCS
	5	May 9, 2024 -23:30h	TCS
	6	May 19, 2024 - 23:30h	EPS
	7	May 28, 2023 - 23:30h	CONF+OBDH
	8a	July 19, 2024 – 18:00h	FINAL REPORT 1st delivery window
	8b	September 9, 2024 – 18:00h	FINAL REPORT 2nd delivery window

Reverse Engineering 1

Missions – Homework 1

- Understand the mission high level goals
- Identify mission drivers, if any
- Perform the mission functional analysis
- Identify its main phases
- Link phases to ConOps and functionalities
- Understand the on board scientific instruments primary utilisation:
 - Correlate goal-to payloads functions
 - Correlate p\l to conops\phases
- Start correlating functionalities-phases trajectory design (MA understanding)

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Reverse Engineering 2

Missions – Homework 2_1

- Understand the mission trajectory design according to:
 - Functionalities and phases
 - ΔV budget per maneuver
- Understand and justify the costs of, through reverse sizing:
 - Main orbital maneuvers → if the mission has complex interplanetary transfer with MGA, select a leg (GA1→GA2 and reconstruct the cost of it)
 - Overall station keeping (center of mass only)
- Reconstruct the DV budget breakdown table with proper margin, as needed

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Missions – Homework 2_2

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- Understand the mission propulsion architecture
- Identify the solutions for **primary and secondary** propulsion and clarify the rationale for the adopted design
- Find the rationale for and justify the selected propulsion type and architecture according to:
 - Operations\phases (which thrusters to do which maneuvering)
 - ΔV budget breakdown
- Understand and justify, through reverse sizing:
 - Propellant selection and masses
 - Tanks sizing (propellant\pressurant), number, material adopted
 - Pressurant selection and masses
 - Feeding strategy selection and sizing
 - # thrusters
 - Positioning of thrusters in the configuration
 - Positioning of tanks and lines in the configuration

Reverse Engineering 3

Missions – Homework 3

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- Understand the mission TMTC architecture
- Identify the solutions for TMTC subsystem and clarify the rationale for the adopted design
- Find the rationale for and justify the selected TMTC type and architecture according to:
 - Operations\phases\modes
 - Data volume to transfer space2ground, ground2space
- Understand and justify, through reverse sizing:
 - Frequency selection, data rate, band
 - Signal manipulation (encoding, modulation, etc)
 - Antenna selection, type, characteristics and numbers correlated to which frequency and for which data transfer
 - Ground station selection: where, which size\frequency\data rate
 - Amplifier selection
 - Contact strategy: contact windows duration and data volume transferred per contact (average)
 - Link budget U/D
 - Positioning of the antennae the configuration on the space segment
 - Subsystem architecture

Reverse Engineering 4

Missions – Homework 4

- Understand the mission AOCS architecture
- Identify the solutions for AOCS subsystem and clarify the rationale for the adopted design
- Find the rationale for and justify the selected AOCS type and architecture according to:
 - Control mode
 - Pointing budget
- Understand and justify, through reverse sizing:
 - Pointing budgets inputs for each subsystem involved: AKE, APE, drift, rates
 - Attitude sensor suite selection according to mode – redundancy included
 - Attitude actuator suite selection according to mode
 - Attitude sensor sizing according to pointing knowledge needs, per mode
 - Disturbances effects, slew maneuvers, per mode\phase
 - Attitude actuator sizing according to disturbances and requested slew maneuvers per mode (torque, angular momentum, thrust, dipole, etc) – redundancy included
 - Fuel mass sizing according to: attitude SK, maneuvers, desaturation, if any
 - Subsystem budgets: mass, power, data
 - Positioning of the sensors and actuators in the s\c configuration

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Reverse Engineering 5

Missions – Homework 5

- Understand the mission TCS solutions
- Identify the solutions applied for the TCS subsystem and clarify the rationale for the adopted design
- Critically motivate the selected TCS architecture according to:
 - External\internal thermal fluxes encountered along the mission phases
 - Requested temperature intervals to be respected on board
- Understand and justify, through reverse sizing:
 - Cold and hot case selection, along the whole mission
 - Adopted control strategy, passive & \or active (e.g. painting, blankets, strips, heaters, etc)
 - Selected materials and areas for passive control
 - Needed electric power for active control, if any
 - Units specifically controlled, if any
 - Subsystem budgets: mass, power, data
 - Positioning of the control components (e.g. surfaces, radiators, heaters, peltier, etc) in the s\c configuration
 - Imposed specific pointing direction for any passive control surface

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Reverse Engineering 6

Missions – Homework 6

- Understand the mission EPS solutions
- Identify the solutions applied for the EPS subsystem and clarify the rationale for the adopted design
- Justify the selected EPS architecture according to:
 - Electrical Power&Energy requested by on board s\s in each mission phases
 - Operational profiles and available sources (i.e. distance from the Sun, Aspect angle,etc)
- Understand and justify, through reverse sizing:
 - Power budget supplied per phase\mode per subsystem
 - Primary source selection and sizing: area\configuration, mass, power demand, #, topology (if applicable), conversion efficiency
 - Secondary source selection and sizing (if any): mass, #, size, energy density, topology, DOD
 - Primary source regulation adopted strategy
 - Bus regulation adopted strategy
 - Subsystem budgets: mass, power, volume, data
 - Positioning of the components in the s\c configuration: PV arrays\wings, battery packages, RTGs
 - specific pointing direction requirements

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Reverse Engineering 7

Missions – Homework 7

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- Understand the configuration of the space segment
- Understand and justify, through reverse sizing:
 - the overall vehicle shape and appendages distribution according to their operational needs and technical requirements
 - The launcher interface location and features and the vehicle configuration when in the launcher fairing
 - the distribution of the elements on the external surface: location, distance\proximity with other components, direction of the FOV, pointing needs, shadowing, etc. according to their operational requisites and constraints
 - the distribution\location of the internal elements with respect to their functionality and operational requisites and constraints: CoM balancing, thermal dissipation

Missions – Homework 7

- Understand the OBDH subsystem desing for the mission
- Identify and describe the adopted architecture in terms of:
 - OBDH components
 - OBDH adopted bus
- Justify the selected OBDH through reverse engineering sizing according to:
 - OBC features as frequency and throughputs by similiarity
 - On board memory size

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